A Preliminary Analysis of Foundation Pose Models on the Quadrotor System.

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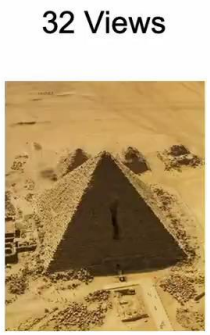
Motivation

- While existing VIO methods (ORB-SLAM, Open-VINS, etc) perform well in controlled environments the have significant weaknesses.
  - Requires inertia data
  - Struggles with high speed flight and dynamic motion
  - Changing lighting conditions
- Objective: We have an existence proof that better VIO is possible, how do we accomplish it ?



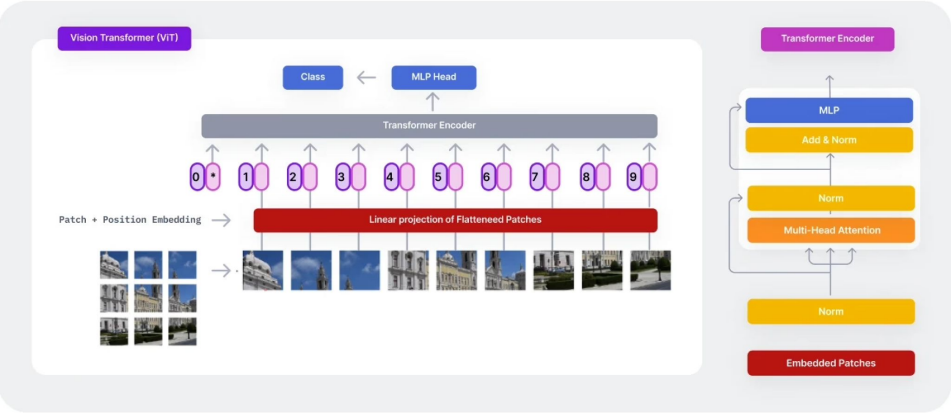
VGGT: Visual Geometry Grounded Transformer

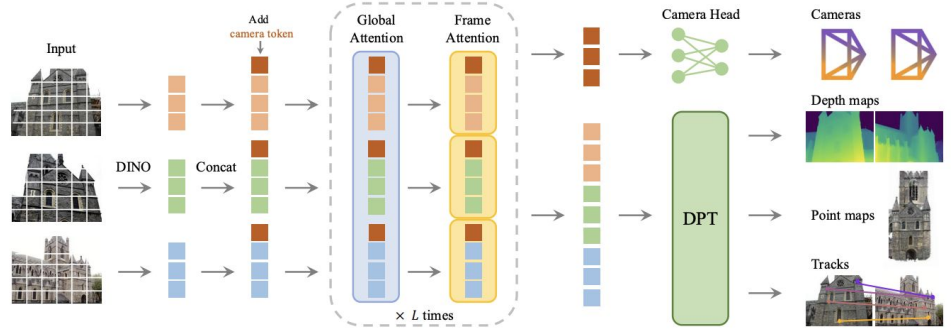
- New learning based approaches to SLAM
  - VGGT, DUSt3R, Fast3R
  - In a single feed forward pass predicts camera extrinsics, intrinsics, depth, PCs, and 2D image correspondences.
- Neural methods have proven to work well on large image batches for reconstruction, but can they would live in the loop?



Understanding ViTs

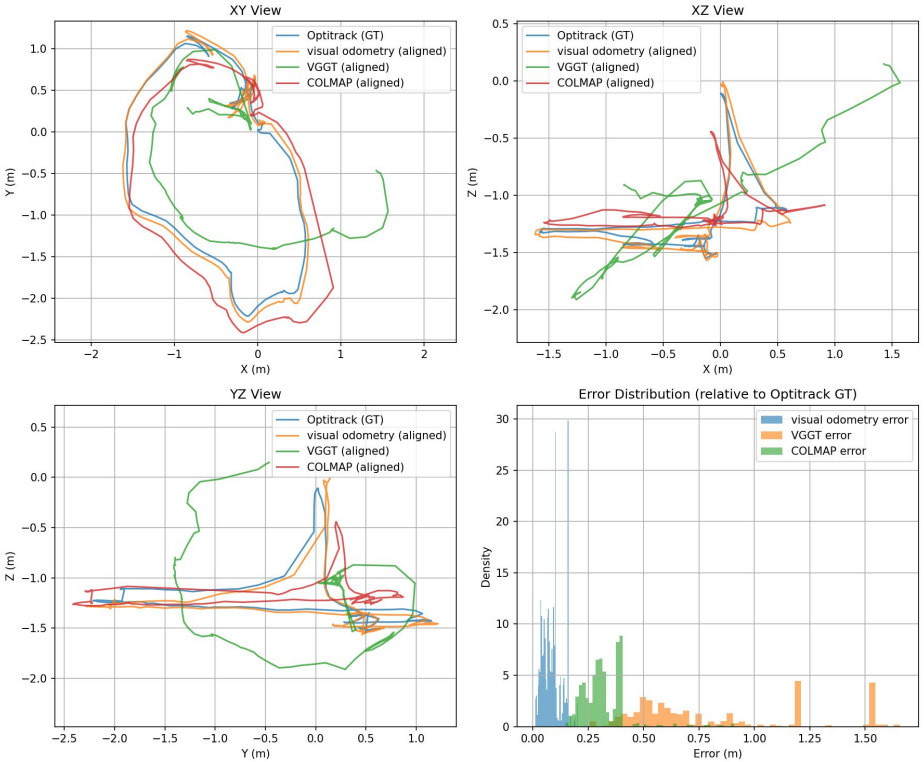
- ViTs chop images into patches, embedding each with positional information, feeding the resulting tokens into the same kind of attention structure that powers ChatGPT
- Patch-wise attention allows the model to reason about relationships between patches, producing globally coherent tokens
- Training such an architecture at scale using self-distillation (DINOv2) allows you to get viewpoint-stable descriptors for any input image without any labels
- Building upon these architectures by fine tuning different heads (DPT, CLS, SAM) allows these powerful models to produce huge amounts of scene information from just a single image





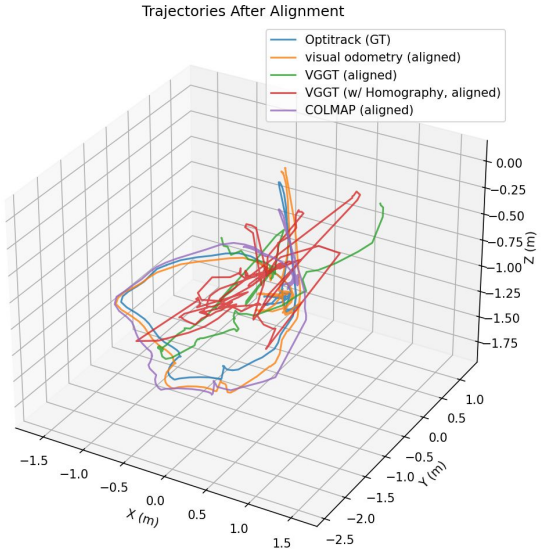
Question: Is VGGT suitable for replacing traditional visual odometry for drone ?

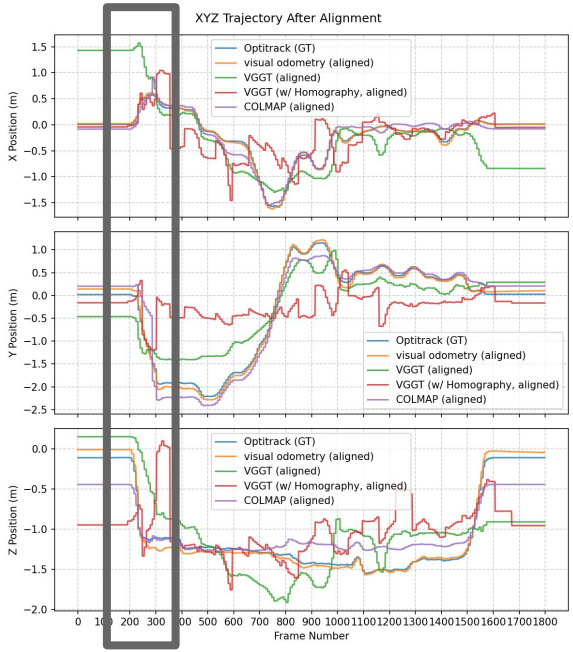
Results - Dynamic “Flight”

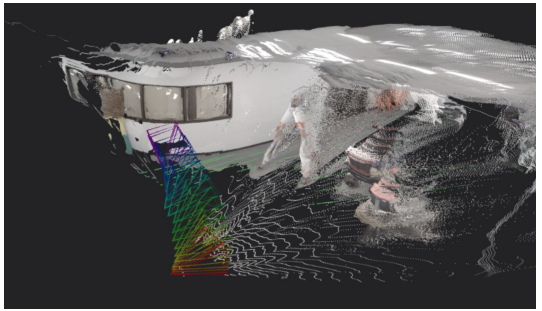


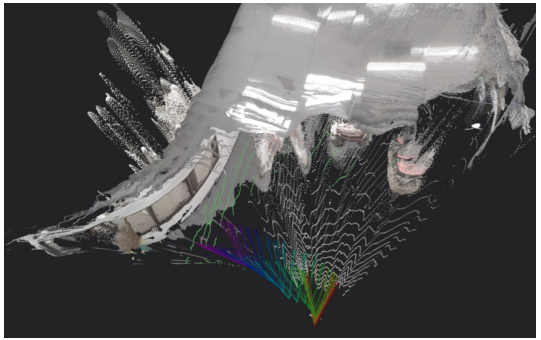


Results - Dynamic “Flight”









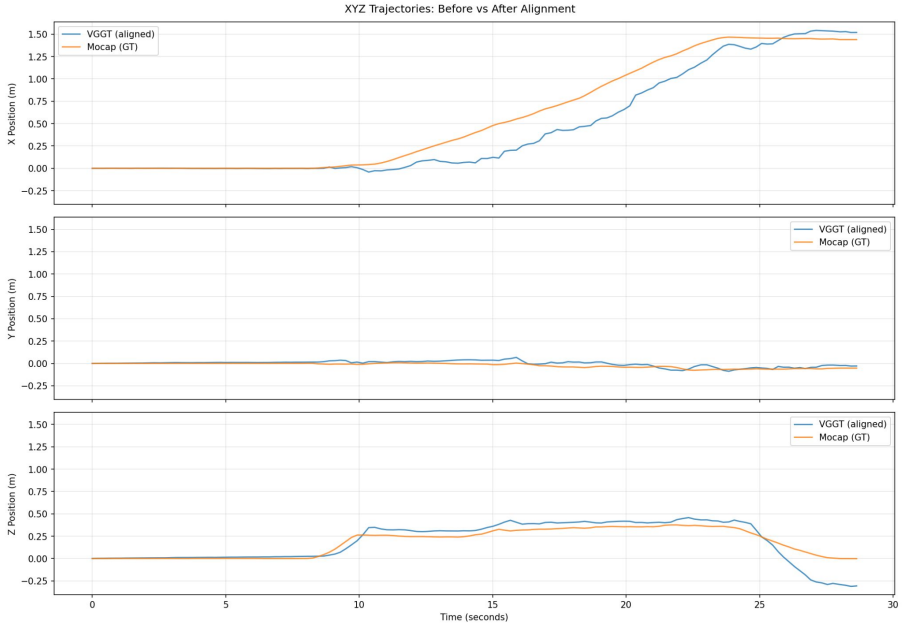
Results - Dynamic “Flight”

Dataset	Total Points	Frequency (Hz)	Data Size
Optitrack	18,726	197.50	(18726, 8)
VIO	2,800	29.55	(2800, 8)
VGGT	463	4.99	(463, 8)
COLMAP	232	2.50	(232, 8)

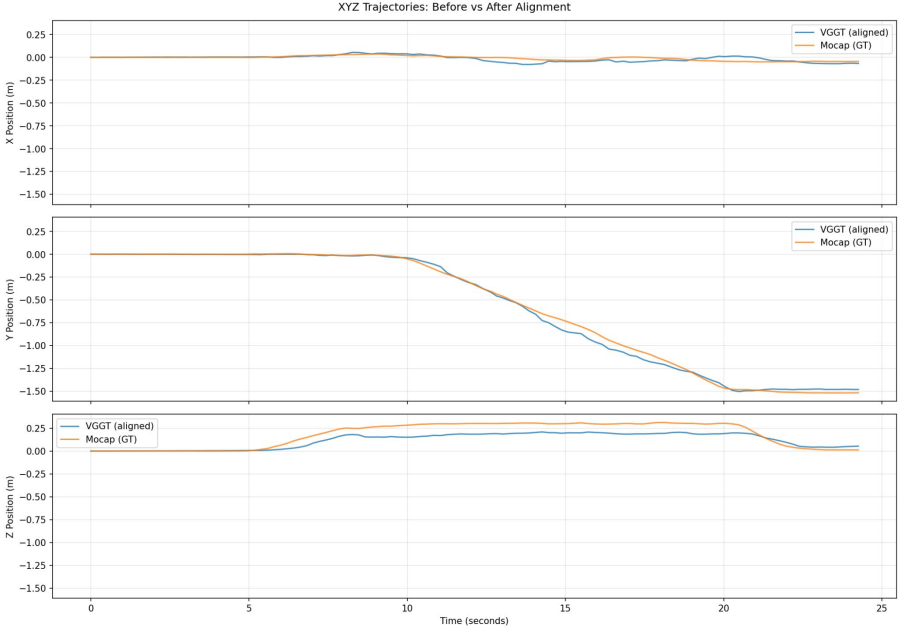
Method	Mean Error (m)	Max Error (m)	Min Error (m)	Std Error (m)
Visual Odometry	0.088	0.204	0.010	0.044
VGGT	0.809	1.657	0.258	0.380
COLMAP	0.317	0.908	0.147	0.100

Results - Simple “Flight”

Move Forward **.15m mean error**

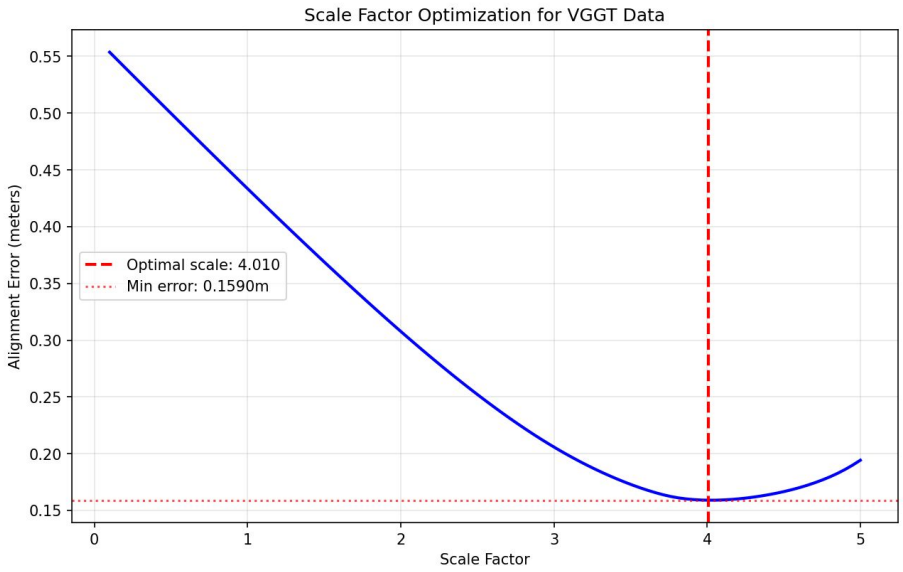


Move Right **.07m mean error**

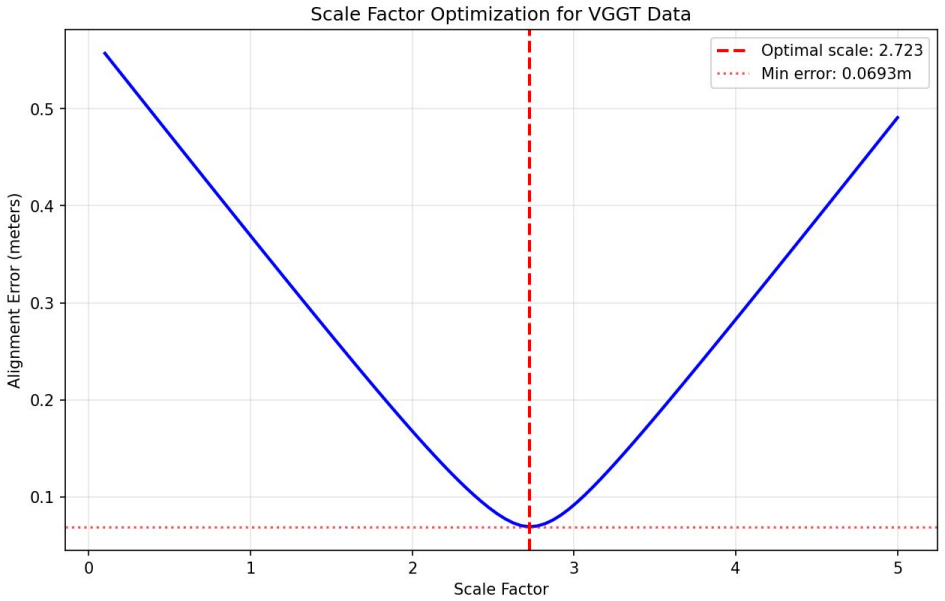


Results - Simple “Flight”

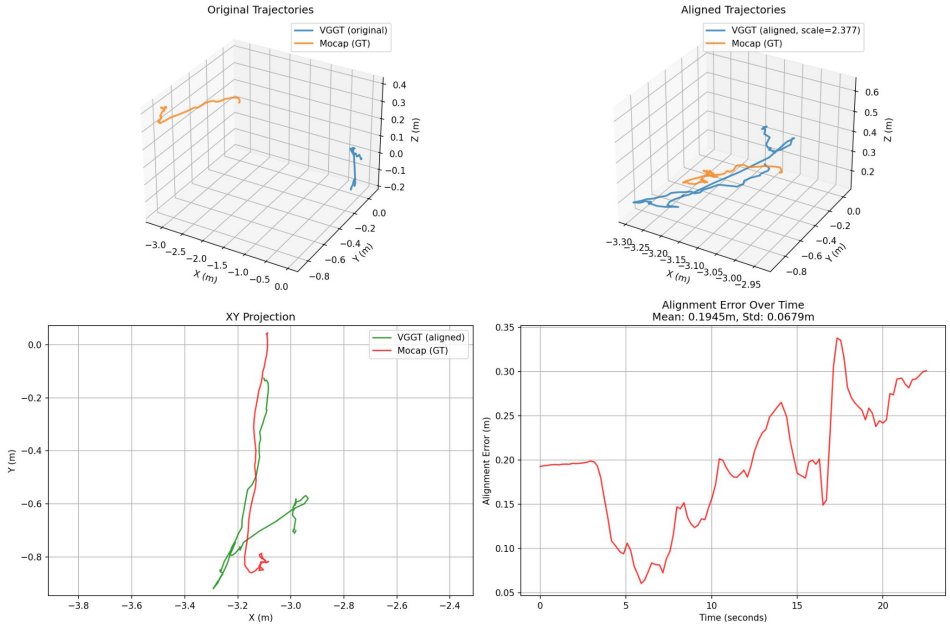
Move Forward **.15m mean error**



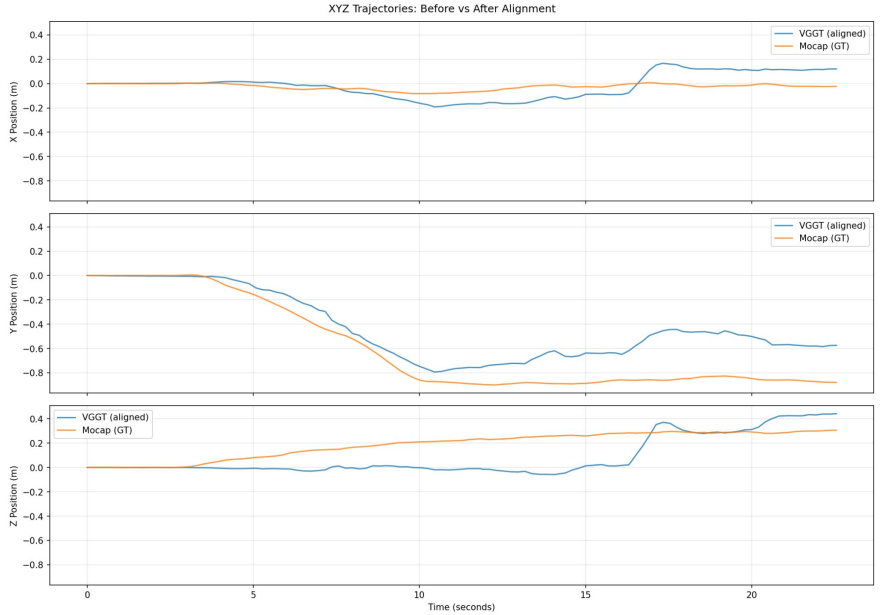
Move Right **.07m mean error**



Results - Medium “Flight”



Move Right Counter Clockwise **.2m mean error**



Conclusions and Next Steps

Conclusions:

- In short and simple trajectories VGGT quantifies motion well.
- In longer and more complex trajectories VGGT struggles.
- VGGT is highly non-metric and exhibits wildly different estimates of scale even across the same scene.

Limitations:

- Struggles with image streams that differ very little from each other
- Lacks the ability to resolve projective ambiguities

Open Questions:

- How to quantify uncertainty with foundation pose models ? (Can we use the PCs)
- Can we architecturally perform loop closure or encode other relations ?
    - Incorporate a recurrent image token descriptor that carries information forward for future reference ?
- Can we improve performance by fine-tuning the model on real world metric data ?
- How do the camera intrinsics affect the model ?